

10 (**long**) Consider the following non-homogeneous second order differential equation

$$\frac{d^2x}{dt^2} + p \frac{dx}{dt} + qx = f(t)$$

where p and q are real coefficients and $f(t)$ is a real function of time t .

(a) For $f(t) = 0$, find the general solution for the following three cases:

(i) $p^2 - 4q > 0$

(ii) $p^2 - 4q = 0$

(iii) $p^2 - 4q < 0$

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(b) Find the specific solution when $f(t) = 2 \sin(t) + \sin(2t)$, $p = 0$, $q = 4$ and $x = dx/dt = 0$ at $t = 0$.

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